Fiscal Year:	FY 2014	**Task Last Updated:**	FY 07/07/2014
PI Name:	Hoffman, Larry Ph.D.		
Project Title:	Gravity-Induced Plasticity in Mammalian Utricular Hair Cells: Intrinsic or Multisensory?		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
PI Email:	lfh@ucla.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	310-825-9823
Organization Name:	University of California Los Angeles		
PI Address 1:	Geffen School of Medicine at UCLA		
PI Address 2:	Box 951624		
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90095-1624	**Congressional District:**	33
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	07/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schweizer, Felix (University of California, Los Angeles) Fuller, Charles (University of California, Davis)		
Grant/Contract No.:	NNX13AL99G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This proposal is being submitted to the research topic area Ground based research designed to lead to space flight announced in NASA NRA NNH12ZTT001N. We are proposing a series of studies to investigate whether the plasticity in the inner ear utricle is affected by non-vestibular load-sensing physiologic systems. Potential interactions between load-sensing systems might offer cross-modal synergy under altered gravity conditions, e.g. loading of the utricle might directly affect other systems (e.g. immune, osteogenic) that are currently subject to significant countermeasure efforts. In addition, the proposed research endeavors to elucidate how this plasticity is manifested in alterations of the activity of neurons that signal such changes to the central nervous system. We plan to use mouse models that systematically exclude the interaction of particular components of the vestibular load-sensing system. These models represent valuable adjuncts that can be implemented in future flight studies to address countermeasures driven by the recruitment of multiple sensory systems. Furthermore, the present investigation serves as a segue between altered gravity exposure		

	durations that have been focused upon the duration of space shuttle missions and the much longer exposures incumbent with experiments aboard the International Space Station.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research explores the natural capabilities of utricular hair cells to undergo modifications in their synaptic connections with neurons that communicate sensory signals to the central nervous system as a result of exposure to altered gravitational fields. The ambient gravitational field is a fundamental component of the natural sensory environment, hence a principal aspect of our research is to explore the "sensory learning" that may result when changes in gravity are imposed. These fundamental findings are the foundation for future experiments to reveal the cellular mechanisms responsible for these changes. Since synaptic decline may reflect one form of natural vestibular dysfunction due to aging, pathology, or ototoxicity, an understanding of this repertoire and the cellular mechanisms is the first step toward designing rehabilitative strategies. Our research is designed to develop a systematic laboratory protocol in which these natural capabilities can be explored.
Task Progress:	Progress toward many of the Aims of our research program has been slow in this first year as we sought to recruit an outstanding postdoctoral scholar to assist in achievement of these Aims. After a long search, we are pleased to report that we have recruited an outstanding young neuroscientist to our research team. This individual comes from a different area of cellular neuroscience, and therefore we are very happy to have recruited a new investigator to space biology research. Despite the delays in recruiting a suitable individual, we made progress on many fronts. This progress will serve the overall progress well once our postdoctoral scholar is able to focus on the Aims. Critical to our Aims is the establishment of two genetically modified mouse strains which enable us to test our hypotheses. Only one of these lines is readily available through a commercial vendor, which will make establishing a colony of this strain at UCLA a readily achievable task. The second line needed to be obtained through the completion of a material transfer agreement with the line inventor, which was successfully executed. Breeding animals were obtained, which initiated the long process of establishing a viable colony (which depended upon successful breeding and care of the progeny, which proved to be a significant challenge initially). After several breedings, we are pleased to report that healthy progeny are being produced driving the process of expanding the colony. During the first grant year we implemented improvements in the technical procedures required to quantify changes in the distribution of synapses in utricular hair cells, representing one outcome measure of synaptic plasticity. These modifications represent a substantial improvement in the elucidation of synapses for quantification, and will eliminate one source of uncertainty in our eventual results in specimens exposed to alterations in the ambient gravity field. We have also implemented novel methods to evaluate the ultrastructure of these synapses, which will provide another metric with which to evaluate changes induced by altered gravity. These techniques, known as serial section field emission scanning electron microscopy, enable us to visualize the entire synaptic structures, rather than small portions as revealed by traditional ultrastructure imaging methods (i.e. electron microscopy). Hence, these methods will expand our analytical repertoire to glean a broader perspective in the types of synaptic modifications that result in utricular hair cells. Our results will not only cast a new light on synaptic changes and sensory learning, but will also provide keen insight into the function of the specialized structure associated with these synapses.
Bibliography Type:	Description: (Last Updated: 10/05/2022)
Abstracts for Journals and Proceedings	Lopez I, Sheets K, Schweizer F, Hoffman L. "Tomographic analyses of afferent synapses in mouse utricular hair cells." Presented at the 37th Annual Midwinter Meeting of the Association for Research in Otolaryngology, San Diego, CA, February 22-26, 2014. Association for Research in Otolaryngology Abstracts. 2014;37:Abstract #PS-218. p. 116-117. , Feb-2014